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# Metasyn: Transparent Generation of Synthetic Tabular Data with Privacy Guarantees

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## Summary

Synthetic data is a promising tool for improving the accessibility of datasets which are too sensitive to be shared publicly. To this end, we introduce `metasyn`, a Python package for generating synthetic data from tabular datasets. Unlike existing synthetic data generation software, `metasyn` is built on a simple generative model that omits multivariate information. This choice enables transparency and auditability, keeps information leakage to a minimum, and enables privacy guarantees through a plug-in system. While the analytical validity of the generated data is thus intentionally limited, its potential uses are broad, including exploratory analyses, code development and testing, and external communication and teaching (van Kesteren, 2024).

**Figure 1:** Logo of the `metasyn` project.

## Statement of need

`Metasyn` is aimed at owners of sensitive datasets such as public organisations, research groups, and individual researchers who want to improve the accessibility of their data for research and reproducibility by others. The goal of `metasyn` is to make it easy for data owners to share the structure and an approximation of the content of their data with others while keeping privacy concerns to a minimum.

With this goal in mind, `metasyn` distinguishes itself from existing software for generating synthetic data (e.g., Nowok et al., 2016; Ping et al., 2017; Templ et al., 2017) by strictly limiting the statistical information from the real data in the synthetic data. `Metasyn` explicitly avoids generating synthetic data with high analytical validity; instead, the synthetic data has realistic structure and plausible values, but multivariate relations are omitted ("augmented plausible synthetic data"; (Bates et al., 2019)). Moreover, our system provides an **auditable and editable intermediate representation** in the form of a `.json` metadata file from which new data can be synthesized.

These choices enable the software to generate synthetic data with **privacy and disclosure**

**guarantees** through a plug-in system, recognizing that different data owners have different needs and definitions of privacy. A data owner can define under which conditions they would accept open distribution of their synthetic data — be it based on differential privacy (Dwork, 2006), statistical disclosure control (Hundepool et al., 2012), k-anonymity (Sweeney, 2002), or another specific definition of privacy. As part of the initial release of `metasyn`, we publish a plug-in following the disclosure control guidelines from Eurostat (Bond et al., 2015).

# Software features

At its core, `metasyn` has three main functions: **estimation**, to fit a model to a properly formatted tabular dataset; **generation**, to synthesize new datasets based on a fitted model; and **(de)serialization**, to create a file from the model for auditing, editing, and saving.

## Estimation

Model estimation starts with an appropriately pre-processed data frame, meaning it is tidy (Wickham, 2014), each column has the correct data type, and missing data are represented by a missing value. Accordingly, `metasyn` is built on the `polars` data frame library (Vink et al., 2024). As an example, the first records of the "hospital" data built into `metasyn` are printed below:

```
┌──────────────┬──────────────┬──────────────┬──────┬──────┬──────────────┐
│ patient_id   ┆ date_admitted ┆ time_admitted ┆ type ┆ age  ┆ hours_in_room │
│ ---          ┆ ---          ┆ ---          ┆ ---  ┆ ---  ┆ ---          │
│ str          ┆ date         ┆ time         ┆ cat  ┆ i64  ┆ f64          │
╞══════════════╪══════════════╪══════════════╪══════╪══════╪══════════════╡
│ A5909X0      ┆ 2024-01-01   ┆ 10:30:00     ┆ IVT  ┆ null ┆ 3.633531     │
│ B4025X2      ┆ 2024-01-01   ┆ 11:23:00     ┆ IVT  ┆ 59   ┆ 6.932891     │
│ B6999X2      ┆ 2024-01-01   ┆ 11:58:00     ┆ IVT  ┆ 77   ┆ 1.970654     │
│ B9525X2      ┆ 2024-01-01   ┆ 16:56:00     ┆ MYE  ┆ null ┆ 1.620047     │
│ …            ┆ …            ┆ …            ┆ …    ┆ …    ┆ …            │
└──────────────┴──────────────┴──────────────┴──────┴──────┴──────────────┘
```

Note that categorical data are encoded as `cat` (not `str`) and missing data is represented by `null` values. Model estimation with `metasyn` is then performed as follows:

```
from metasyn import MetaFrame
mf = MetaFrame.fit_dataframe(df_hospital)
```

The generative model in `metasyn` makes the simplifying assumption of *marginal independence*: each column is considered separately, similar to naïve Bayes classifiers (Hastie et al., 2009). For each column, a set of candidate distributions is fitted (see Table 1), and then `metasyn` selects the one that fits best (usually having the lowest BIC (Neath & Cavanaugh, 2012)). Key advantages of this approach are transparency and explainability, flexibility in handling mixed data types, and computational scalability to high-dimensional datasets.

**Table 1:** Candidate distributions associated with data types in the core `metasyn` package.

| Data type | Candidate distributions |
|---|---|
| Categorical | Categorical, Constant |
| Continuous | Uniform, Normal, LogNormal, TruncatedNormal, Exponential, Constant |
| Discrete | Poisson, Uniform, Normal, TruncatedNormal, Categorical, Constant |
| String | Regex, Categorical, Faker, FreeText, Constant |
| Date/time | Uniform, Constant |

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From this table, the string distributions deserve special attention as they are not common probability distributions. The regex (regular expression) distribution uses the package regexmodel to automatically detect structure such as room numbers (A108, C122, B109), identifiers, e-mail addresses, or websites. The FreeText distribution detects the language (using lingua) and randomly picks words from that language. The Faker distribution can generate specific data types such as localized names and addresses pre-specified by the user.

## Data generation

After creating a `MetaFrame`, `metasyn` can randomly sample synthetic datapoints from it. This is done using the `synthesize()` method:

```
df_syn = mf.synthesize(3)
```

This may result in the following data frame. Note that missing values in the age column are appropriately reproduced as well.

```
┌──────────────┬──────────────┬──────────────┬──────┬──────┬──────────────┐
│ patient_id   ┆ date_admitted ┆ time_admitted ┆ type ┆ age  ┆ hours_in_room │
│ ---          ┆ ---          ┆ ---          ┆ ---  ┆ ---  ┆ ---          │
│ str          ┆ date         ┆ time         ┆ cat  ┆ i64  ┆ f64          │
╞══════════════╪══════════════╪══════════════╪══════╪══════╪══════════════╡
│ B7906X1      ┆ 2024-01-04   ┆ 13:32:00     ┆ IVT  ┆ 37   ┆ 4.955418     │
│ B0553X2      ┆ 2024-01-02   ┆ 10:54:00     ┆ IVT  ┆ 39   ┆ 3.872872     │
│ A5397X7      ┆ 2024-01-03   ┆ 18:16:00     ┆ CAT  ┆ null ┆ 6.569082     │
└──────────────┴──────────────┴──────────────┴──────┴──────┴──────────────┘
```

## Serialization and deserialization

`MetaFrames` can also be transparently stored in a human- and machine-readable `.json` metadata file. This file contains dataset-level descriptive information as well as variable-level information. This `.json` can be manually audited, edited, and after saving this file, an unlimited number of synthetic records can be created without incurring additional privacy risks. Serialization and deserialization with `metasyn` is done using the `save()` and `load()` methods:

```
mf.save("hospital_admissions.json")
mf_new = MetaFrame.load("hospital_admissions.json")
```

# Privacy

As a general principle, `metasyn` errs on the side of privacy by default, aiming to recreate the structure but not all content and relations in the source data. For example, take the following sensitive dataset where study participants state how they use drugs in daily life:

```
┌──────────────┬──────────────────────────────┐
│ participant_id ┆ drug_use                     │
│ ---          ┆ ---                          │
│ str          ┆ str                          │
╞══════════════╪══════════════════════════════╡
│ OOWJAHA4     ┆ I use marijuana in the evening… │
│ 8CA1RV4P     ┆ I occasionally take CBD to hel… │
│ FMSVAKPM     ┆ Prescription medication helps … │
│ …            ┆ …                            │
└──────────────┴──────────────────────────────┘
```

When creating synthetic data for this example, the information in the open answers is removed, and using our standard `FreeText` distribution this information is replaced by words from the

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detected language (English):

```
┌──────────────┬──────────────────────────────┐
│ participant_id ┆ drug_use                     │
│ ---          ┆ ---                          │
│ str          ┆ str                          │
╞══════════════╪══════════════════════════════╡
│ ZQJZQAB7     ┆ Lawyer let sort her yet line e… │
│ 7KDLEL0S     ┆ Particularly third myself edge… │
│ QBZKGXC7     ┆ Put color against call researc… │
└──────────────┴──────────────────────────────┘
```

Additionally, the `metasyn` package supports plug-ins which alter the estimation behaviour. Through this system, privacy guarantees can be built into `metasyn` and additional distributions can be supported. For example, `metasyn-disclosure-control` implements output guidelines from Eurostat (Bond et al., 2015) through *micro-aggregation*.


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